OptiLife — AI Lifestyle Optimizer

Project Documentation | DEPI – Digital Egypt Pioneers Initiative | Track: Cross Platform Mobile App

Repository: https://github.com/mohamedkiilany/OptiLife

Stage: Project Planning & Management, Literature Review, Requirements Gathering

Items marked "Proposed" are suggestions to be reviewed and confirmed with the team and lecturer.

Contents

1. Project Planning & Management
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1. Project Planning & Management

1.1 Project Idea

Overview. OptiLife (AI Lifestyle Optimizer) is an AI-powered personal wellness application that helps users build healthier lifestyles based on their real-world constraints. It combines fitness, nutrition, time and budget, and adapts its recommendations to what the user actually has instead of assuming enough time, money, equipment or gym access.

Core question: "What is the healthiest lifestyle I can realistically follow with the time, money, and resources I have?"

Problem. Many fitness and nutrition apps assume users have enough time to exercise, a gym, a sufficient food budget, access to the recommended ingredients, workout equipment and time to prepare complex meals. Real users often have only 20 minutes, a limited budget, no equipment, limited gym access or few ingredients at home, so ideal plans are hard to follow.

Proposed solution. The app generates a personalized plan that combines fitness and nutrition. The user defines:

Input	Options
Goal	Lose Weight, Gain Muscle, Maintain Weight, Improve Fitness
Available time	10 minutes, 20 minutes, 30 minutes, 45+ minutes
Training environment	Home, Gym, Outdoors
Equipment	None, Dumbbells, Resistance Bands, Gym Equipment
Food budget	Low, Medium, High

Example. A user says: "I want to lose weight, I only have 20 minutes per day, I don't go to the gym, and I have a limited food budget." The app generates a 20-minute home workout, a low-cost meal plan, a shopping list of affordable ingredients, lower-cost alternatives for expensive ingredients, and a realistic daily schedule for workouts and meals.

AI integration. The AI uses the user's goal, available time, budget, equipment, food availability, dietary preferences and fitness level to generate personalized recommendations. The AI is not just a chatbot; it acts as the personalization and decision-making layer of the application.

1.2 Objectives

- Generate fitness and nutrition plans that fit the user's time, budget, equipment and food availability.
- Use AI as the personalization and decision-making layer, not only as a chatbot.
- Let users track fitness and nutrition progress over time.
- Deliver the app as a cross-platform mobile application using Flutter.

1.3 Scope

In scope (core features):

- Personalized user profile
- AI workout planner and AI meal planner
- Budget-friendly food alternatives
- "What Can I Make?" ingredient-based meal suggestions
- "What Can I Do Today?" time-based workout suggestions
- Smart daily planning
- AI wellness assistant

- Fitness and nutrition progress tracking

Out of scope (Proposed): medical diagnosis or treatment advice, integration with wearable devices, online payments or grocery ordering, and social or community features.

1.4 Unique Value Proposition

OptiLife does not give users an ideal lifestyle. It creates a healthier lifestyle they can actually afford and follow. Instead of asking "What is the perfect diet and workout plan?", the app asks "What is the best plan for this person, given their actual time, budget, and resources?"

1.5 Project Plan

Milestones follow the DEPI deadlines from the Project Documentation Guidelines. Start dates in the schedule are proposed and can be adjusted.

Milestone	DEPI deadline	Main deliverables
Project Planning & Management	16 Oct 2026	Proposal, plan, roles, risks, KPIs
Literature Review	16 Oct 2026	Similar solutions, comparison, references
Requirements Gathering	16 Oct 2026	Stakeholders, user stories, use cases, requirements
System Analysis & Design	6 Nov 2026	Use case, architecture, ERD, schema, DFD, UML, wireframes, tech stack
Implementation	30 Nov 2026	Source code, GitHub repo, README, executable app
Final Presentation, Testing & Reports	4 Dec 2026	Test plan, user manual, technical docs, presentation, infographic

Proposed schedule (weekly view)

Activity	Week 1 (5 Oct)	Week 2 (12 Oct)	Week 3 (19 Oct)	Week 4 (26 Oct)	Week 5 (2 Nov)	Week 6 (9 Nov)	Week 7 (16 Nov)	Week 8 (23 Nov)	Week 9 (30 Nov)
Planning & Management	■	■							
Literature Review	■	■							

Activity	Week 1 (5 Oct)	Week 2 (12 Oct)	Week 3 (19 Oct)	Week 4 (26 Oct)	Week 5 (2 Nov)	Week 6 (9 Nov)	Week 7 (16 Nov)	Week 8 (23 Nov)	Week 9 (30 Nov)
Requirements Gathering		■							
System Analysis & Design			■	■	■				
Implementation					■	■	■	■	
Testing & QA							■	■	■
Final Presentation & Reports								■	■

Resource allocation

Resource	Details
Frontend	Flutter / Dart, Bloc / Cubit state management
Backend and data	Firebase, Cloud Firestore, Firebase Authentication
Networking	Dio with interceptors
AI and data services	AI API, external nutrition APIs
Local storage	Hive / SharedPreferences
Testing	Unit tests, widget tests
Version control	GitHub (all documents are uploaded there)

1.6 Task Assignment & Roles

The work is divided by **feature**. Because the whole team works in Flutter, each member owns one part of the app end to end (screens, Bloc / Cubit state, data and tests). Firebase, the AI API and the nutrition APIs are connected to each feature through a shared core, so no one has to build a whole system alone.

Team members:

- Mohamed Ibrahim Abdelkareem Kilany — Team Leader
- Marwan Mohamed Kassem Mohamed
- Thuraya Sami Abdelaty Abu Shuaisha
- Esraa Osama Hassan Abdelaal
- Rawan El-Sayed Ahmed Radwan

- Yassin Ahmed Abdelraouf Yassin

#	Feature area	Responsibilities	Assigned to
1	Authentication & Profile (also Project lead)	Login, Sign Up, User Profile; builds the shared core first (Firebase setup, Dio and interceptors, theme, routing, base AI service); plans milestones, manages GitHub and reviews pull requests	Mohamed Ibrahim Abdelkareem Kilany
2	Home Dashboard & Smart Daily Schedule	Home dashboard and the combined daily schedule of workouts and meals; starts with mock data, then connects to the real plans	Rawan El-Sayed Ahmed Radwan
3	AI Workout Planner & "What Can I Do Today?"	Workout plan generation and time-based workout suggestions	Marwan Mohamed Kassem Mohamed
4	AI Meal Planner, Shopping List & Budget Alternatives	Meal plan generation, shopping list, cheaper ingredient alternatives	Esraa Osama Hassan Abdelaal
5	"What Can I Make?", Nutrition APIs & Progress Tracking	Ingredient-based meal suggestions, nutrition API integration, fitness and nutrition progress tracking	Thuraya Sami Abdelaty Abu Shuaisha
6	AI Wellness Assistant	Chat assistant that gives personalized advice using the user's profile and plans	Yassin Ahmed Abdelraouf Yassin

Shared responsibilities: each member designs the screens, writes the unit and widget tests, and documents the feature they own. Thuraya coordinates the final documentation and user manual.

1.7 Risk Assessment & Mitigation Plan

All risks below are proposed and based on the project's described features and stack.

#	Risk	Impact	Mitigation
R1	AI generates unsuitable or unsafe workout or meal advice	High	Constrain prompts to the user's inputs, add a visible wellness disclaimer, review sample outputs during testing
R2	AI API errors, latency or usage limits	High	Timeouts and retries in Dio interceptors, friendly error states, cache the last plan locally
R3	External nutrition API is unavailable or returns incomplete data	Medium	Handle failures gracefully, fall back to cached data
R4	User health and profile data privacy	High	Firebase Authentication, Firestore security rules, collect only the data needed
R5	Schedule pressure before the 30 Nov and 4 Dec deadlines	Medium	Prioritize core features first, track milestones weekly
R6	Team member unavailable or skills gap	Medium	Share knowledge, keep tasks small, use pull requests and code review
R7	Firebase or API usage costs exceed budget	Low	Monitor usage, use free tiers during development

1.8 KPIs (Key Performance Indicators)

These KPIs are proposed. Confirm or change them with the team and lecturer.

KPI	How it is measured
Plan generation response time	Time from request to displayed plan
System availability	Backend and AI API uptime during testing
Constraint match rate	Share of generated plans that respect the chosen time, budget and equipment in testing
Test coverage and pass rate	Unit and widget tests passing
Crash-free sessions	Sessions without a crash in testing
User adoption	Number of testers who complete a profile and generate a plan

2. Literature Review

2.1 Purpose and Method

This section studies existing fitness, nutrition and wellness applications that solve a problem similar to OptiLife's. For each solution we review what it offers, its strengths and weaknesses, and how it compares with OptiLife. The findings were gathered from each product's official website and app-store description. Feature and pricing details change often, so they should be re-checked against the sources in Section 2.7 before final submission.

2.2 Similar Solutions

Solution	Main focus	Strengths	Weaknesses (relative to OptiLife's goal)
MyFitnessPal	Calorie and food logging	Very large food database, barcode scanning, long-established tracking tools	Focused on logging, not on building a plan around the user's time and budget; several advanced features are behind a paid plan
Fitbod	Strength workout generation	Generates workouts from the user's equipment, goals and training history	Training only, with no meal planning or food budget; subscription-based

Solution	Main focus	Strengths	Weaknesses (relative to OptiLife's goal)
Freeletics	Bodyweight training with AI coaching	Works without equipment, adaptive training programs, optional nutrition guidance	Mostly subscription-based; does not plan meals around the user's budget or the ingredients at home
Nike Training Club	Guided workout library	Large library of workouts at many lengths, strong video guidance	Library browsing rather than a personalized plan; no nutrition or budget awareness
Mealime	Meal planning and grocery lists	Simple meal plans and automatic shopping lists	Nutrition only, with no workout planning; limited awareness of the user's time for exercise
Noom	Behavior-based weight loss	Coaching and psychology-based lessons, daily habit support	Costly subscription, mainly weight-loss oriented, not designed around the user's equipment or ingredient constraints
Cronometer	Detailed nutrient tracking	Detailed tracking of calories and micronutrients	Tracking tool; no workout planner and no budget-based suggestions

2.3 What Existing Solutions Do Well

- **Large food databases and easy logging** (MyFitnessPal, Cronometer) make tracking practical.
- **Equipment-aware workout generation** (Fitbod, Freeletics) shows that workouts can adapt to the user's setup.
- **Automatic shopping lists** (Mealime) save planning time and reduce food waste.
- **Habit and coaching features** (Noom) improve long-term engagement.
- **Short, varied workout formats** (Nike Training Club) fit different schedules.

2.4 Gaps in Existing Solutions

- Most apps address **one area** only (workouts, meals or tracking), so the user must combine several apps.
- **Food budget** is rarely a primary input, and cheaper substitutes for expensive ingredients are uncommon.
- **Time is usually a filter**, not the central constraint used to build the entire day's plan.
- Few apps suggest meals from the **ingredients the user already has** while also considering the fitness goal.
- Many of the most useful features require a **paid subscription**, which conflicts with users who have a limited budget.

2.5 Comparison with OptiLife

Capability	MyFitnessPal	Fitbod	Freeletics	Nike Training Club	Mealime	Noom	OptiLife
Workout planning	No	Yes	Yes	Library	No	Limited	Yes
Meal planning	Limited	No	Limited	No	Yes	Limited	Yes
Food budget as an input	No	No	No	No	Limited	No	Yes
Cheaper ingredient alternatives	No	No	No	No	No	No	Yes
Ingredient-based suggestions	No	No	No	No	Limited	No	Yes
Time-based workout suggestions	No	Limited	Yes	Yes	No	No	Yes
Combined daily schedule (fitness + nutrition)	No	No	No	No	No	No	Yes
AI as the decision-making layer	Limited	Yes	Yes	No	No	Limited	Yes

The comparison is based on the main advertised features of each product and should be confirmed against the references.

2.6 Lessons Applied to OptiLife

- Keep the **input simple** (selection-based choices), as the strongest apps do, so a plan is created in a few taps.
- Use **equipment and environment awareness** for workouts, as Fitbod and Freeletics do.

- Include an **automatic shopping list**, as Mealime does, and extend it with budget alternatives.
- Add **progress tracking** to support engagement, as Noom and MyFitnessPal do.
- Show a **clear wellness disclaimer** and constrain AI output to the user's inputs, since AI-generated health guidance can be unsafe if unchecked.
- Make the core value available without a hard paywall, since the target users are budget-constrained.

2.7 References

1. MyFitnessPal — https://www.myfitnesspal.com
2. Fitbod — https://fitbod.me
3. Freeletics — https://www.freeletics.com
4. Nike Training Club — https://www.nike.com/ntc-app
5. Mealime — https://www.mealime.com
6. Noom — https://www.noom.com
7. Cronometer — https://cronometer.com
8. Flutter documentation — https://docs.flutter.dev
9. Firebase documentation — https://firebase.google.com/docs
10. Bloc library documentation — https://bloclibrary.dev

Add the access date next to each source when you finalize the document.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholders are identified from the project description. Confirm and extend them with the team.

Stakeholder	Needs
End users (people with limited time, budget, equipment or gym access)	Realistic, affordable, personalized fitness and nutrition plans they can actually follow; simple input; progress tracking
Development team	Clear requirements, a maintainable Flutter / Firebase codebase, manageable scope
DEPI lecturer / evaluators	Complete documentation, working implementation, testing and presentation

Stakeholder	Needs
Third-party providers (AI API, nutrition APIs, Firebase)	Usage within their terms, limits and keys

3.2 User Stories

- As a user, I want to set my goal, available time, environment, equipment and food budget so that my plan fits my life.
- As a user with only 20 minutes a day, I want a short home workout so that I can exercise without a gym.
- As a user with a limited budget, I want a low-cost meal plan, a shopping list and cheaper alternatives so that I can eat healthier affordably.
- As a user, I want to enter the ingredients I have so that the app suggests what I can make ("What Can I Make?").
- As a user, I want workout suggestions based on the time I have today so that I can stay active ("What Can I Do Today?").
- As a user, I want a daily schedule for workouts and meals so that I know what to do and when.
- As a user, I want to ask the AI wellness assistant questions so that I get guidance personalized to my situation.
- As a user, I want to track my fitness and nutrition progress so that I can see my improvement.

3.3 Use Cases

ID	Use case	Actor	Description
UC1	Register / log in	User	Create an account and sign in with Firebase Authentication
UC2	Create / update profile	User	Set goal, time, environment, equipment, food budget, dietary preferences, fitness level
UC3	Generate workout plan	User, AI	AI creates a workout matching time, environment and equipment
UC4	Generate meal plan	User, AI, Nutrition API	AI creates a meal plan, shopping list and cheaper alternatives for the budget

ID	Use case	Actor	Description
UC5	What Can I Make?	User, AI	User lists available ingredients and receives meal suggestions
UC6	What Can I Do Today?	User, AI	User picks today's available time and receives workout suggestions
UC7	View daily plan	User	See a combined schedule of workouts and meals
UC8	Ask wellness assistant	User, AI	Chat with the AI assistant for personalized advice
UC9	Track progress	User	Log and view fitness and nutrition progress

3.4 Functional Requirements

ID	Requirement	Source
FR1	The system shall let users register, log in and log out using Firebase Authentication.	Proposed
FR2	The system shall store a personalized user profile (goal, available time, training environment, equipment, food budget, dietary preferences, fitness level).	Stated
FR3	The system shall generate a personalized workout plan based on the user's goal, time, environment, equipment and fitness level.	Stated
FR4	The system shall generate a personalized meal plan based on goal, budget, food availability and dietary preferences.	Stated
FR5	The system shall provide budget-friendly alternatives for expensive ingredients and a shopping list of affordable ingredients.	Stated
FR6	The system shall suggest meals from ingredients the user already has ("What Can I Make?").	Stated
FR7	The system shall suggest workouts for the time available today ("What Can I Do Today?").	Stated

ID	Requirement	Source
FR8	The system shall produce a smart daily schedule combining workouts and meals.	Stated
FR9	The system shall provide an AI wellness assistant for personalized questions.	Stated
FR10	The system shall let users track fitness and nutrition progress.	Stated
FR11	The system shall retrieve nutrition data from external nutrition APIs.	Stated
FR12	The system shall store user data in Cloud Firestore and keep local data using Hive / SharedPreferences.	Stated

3.5 Non-functional Requirements

The technology choices are stated in the project description. The quality requirements are proposed; adjust them to match the KPIs.

Category	Requirement	Source
Performance	Plans and suggestions should appear within an acceptable time, with loading indicators.	Proposed
Security	Users must authenticate; Firestore access limited to the owner's data; API keys kept out of the app source and repository.	Proposed
Usability	Simple selection-based inputs (time, budget, equipment) so a plan can be created in a few taps.	Proposed
Reliability	Graceful handling of AI and nutrition API failures; last plan available offline from local storage.	Proposed
Compatibility	Single Flutter / Dart codebase for cross-platform mobile.	Stated
Maintainability	Bloc / Cubit state management, Dio with interceptors, unit and widget tests.	Stated
Safety	AI-generated health guidance shows a clear wellness disclaimer.	Proposed

3.6 Assumptions and Constraints (Proposed)

- Users have a smartphone with internet access for AI plan generation.
- AI and nutrition API usage stays within free or low-cost tiers during development.
- The project must meet the DEPI deadlines listed in Section 1.5.
- The app provides general wellness guidance and does not replace professional medical advice.